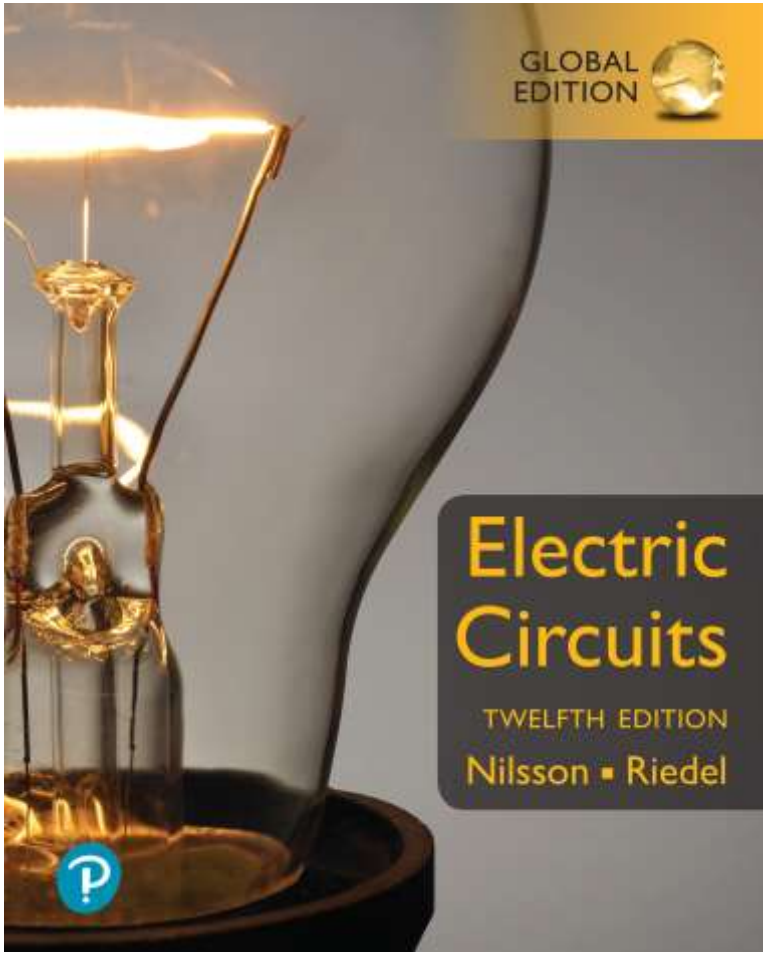
Electric Circuits

Twelfth Edition, Global Edition



Chapter 3 – Part 1

Simple Resistive Circuits

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Chapter 3 – Simple resistive circuits
(1 of 20)

Objectives:

3.1 Know how to identify components connected in series and in parallel

3.2 Be able to find the equivalent resistance of series-connected and parallel-connected resistors

3.3 Be able to recognize circuits where voltage division occurs and know how to use voltage division to analyze such circuits

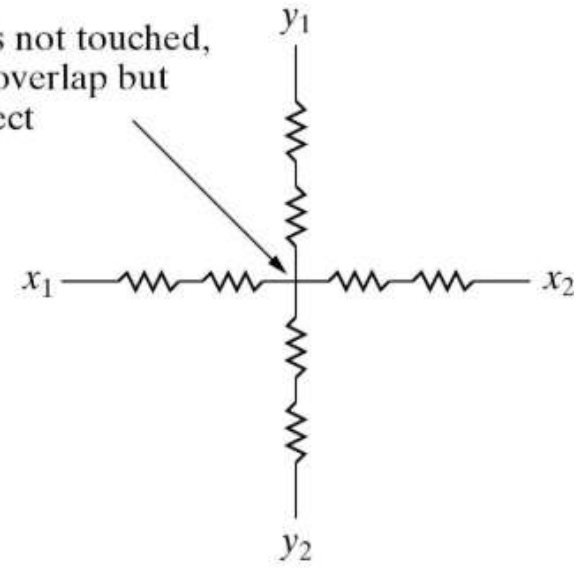
3.4 Be able to recognize circuits where current division occurs and know how to use current division to analyze such circuits

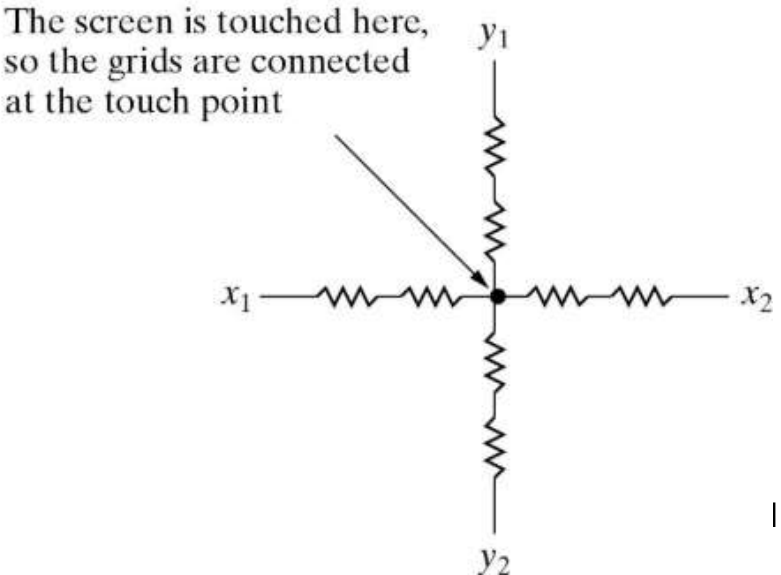
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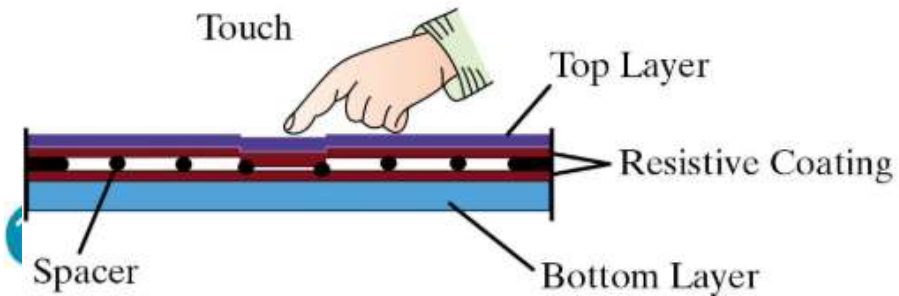
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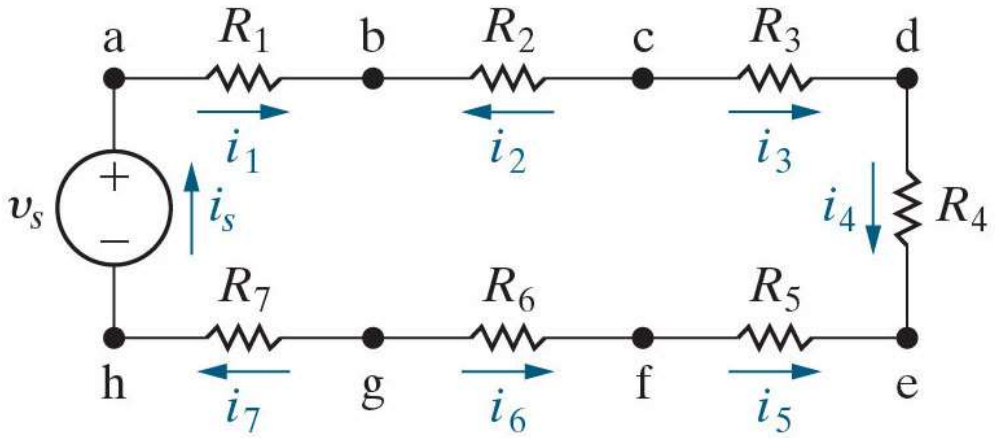




Chapter 3 – Simple resistive circuits
(2 of 20)

Resistors in series:

- Connected end-to-end
- Have the same current (because of KCL)



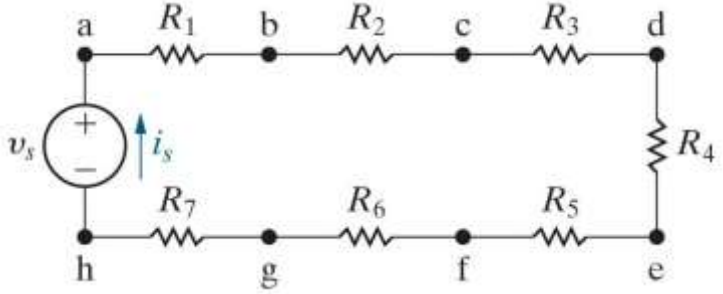
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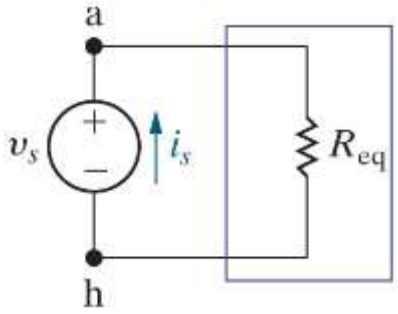
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Chapter 3 – Simple resistive circuits
(3 of 20)

Resistors in series:

- Find the equivalent resistance





KVL: $-v_s + R_1 i_s + R_2 i_s + \cdots + R_7 i_s = 0$

$\Rightarrow \quad v_s = (R_1 + R_2 + \cdots + R_7) i_s$

KVL: $-v_s + R_{eq} i_s = 0$

$\Rightarrow \quad v_s = (R_{eq}) i_s$

Resistors in series add: $R_{eq} = (R_1 + R_2 + \cdots + R_7)$

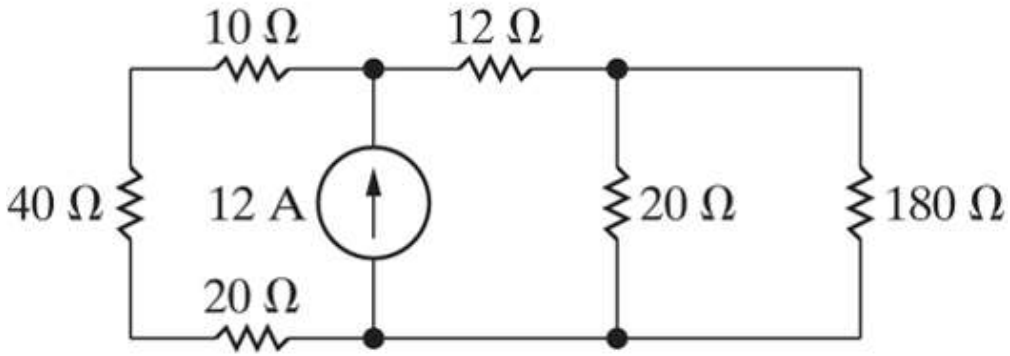
Equivalent resistance is always larger than the largest resistance in the group.

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Chapter 3 – Simple resistive circuits
(4 of 20)

Which of the resistors in the circuit below are connected in series?



A. 10Ω and 12Ω

B. 20Ω and 180Ω

C. 10Ω and 40Ω and 20Ω

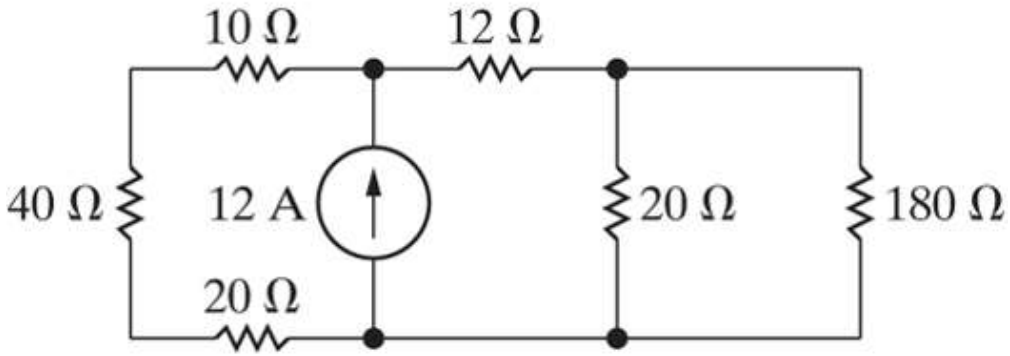
D. 12Ω and 180Ω

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Chapter 3 – Simple resistive circuits
(4 of 20)

Which of the resistors in the circuit below are connected in series?



✗ A. 10Ω and 12Ω

✗ B. 20Ω and 180Ω

✓ C. 10Ω and 40Ω and 20Ω

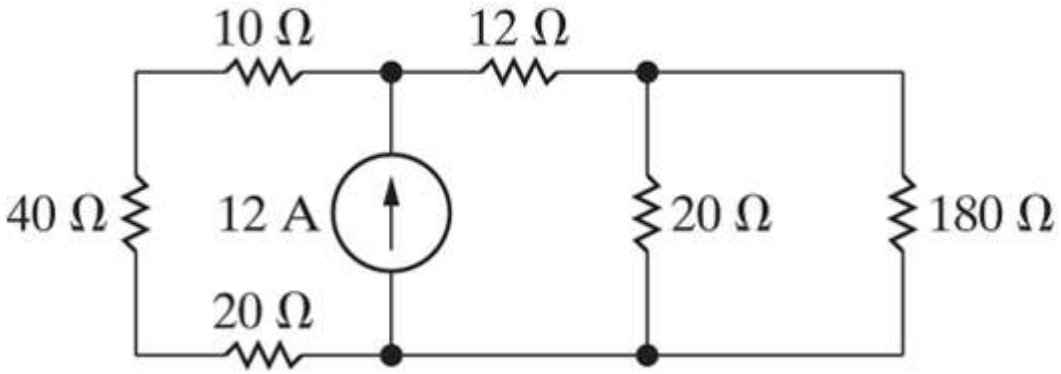
✗ D. 12Ω and 180Ω

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Chapter 3 – Simple resistive circuits
(5 of 20)

Find the equivalent resistance of the series-connected resistors in the circuit here



A. 70Ω

B. 200Ω

C. 82Ω

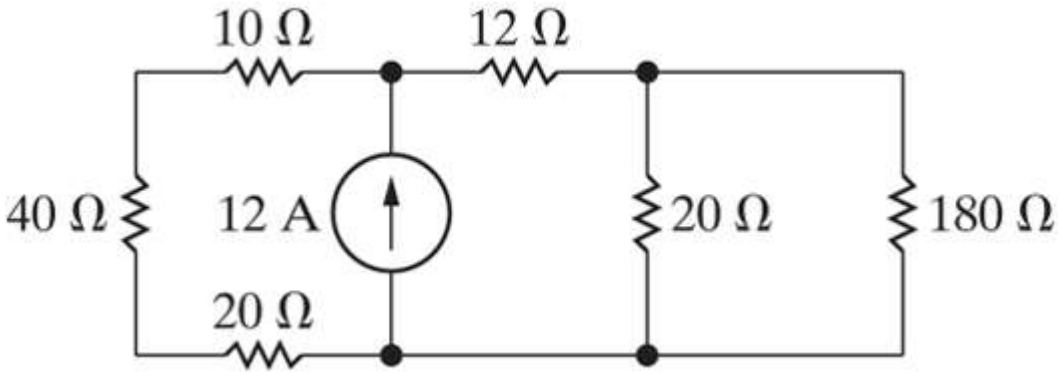
D. 18Ω

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Chapter 3 – Simple resistive circuits
(5 of 20)

Find the equivalent resistance of the series-connected resistors in the circuit here



- ✔ A. 70Ω
- ✘ B. 200Ω
- ✘ C. 82Ω
- ✘ D. 18Ω

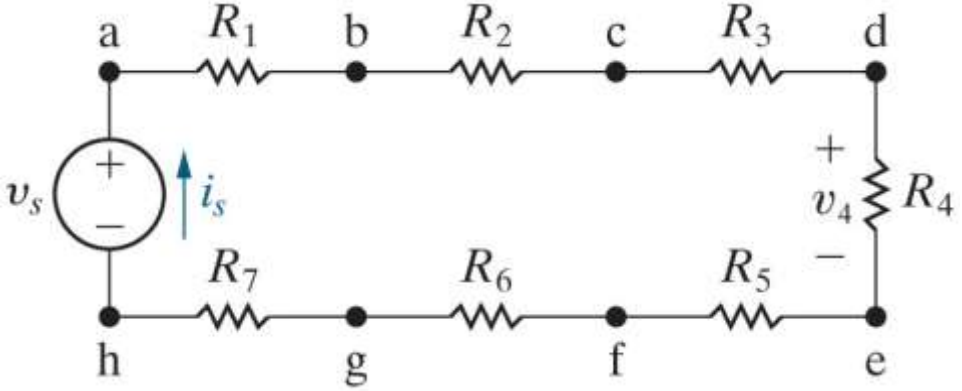
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Chapter 3 – Simple resistive circuits
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Voltage division:

- Find the voltage drop across one of a group of series-connected resistors



$$v_4 = R_4 i_s = R_4 \frac{v_s}{(R_1 + R_2 + \cdots + R_7)} = \frac{R_4}{R_{eq}} v_s$$

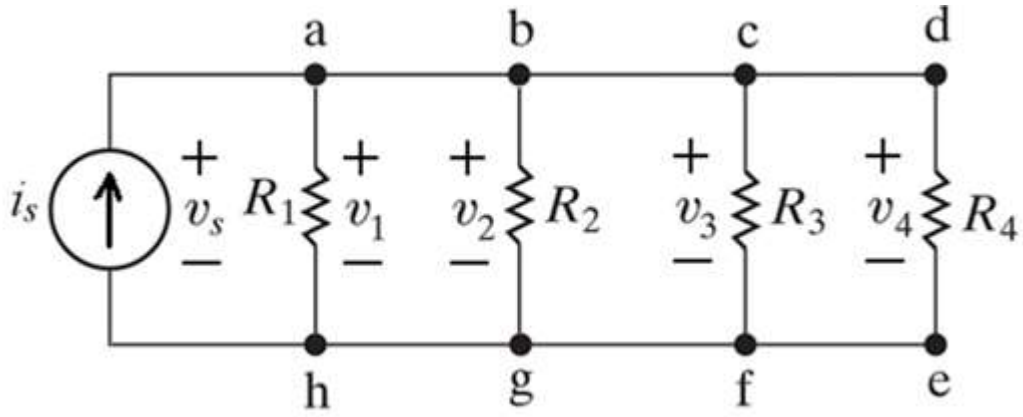
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Chapter 3 – Simple resistive circuits
(8 of 20)

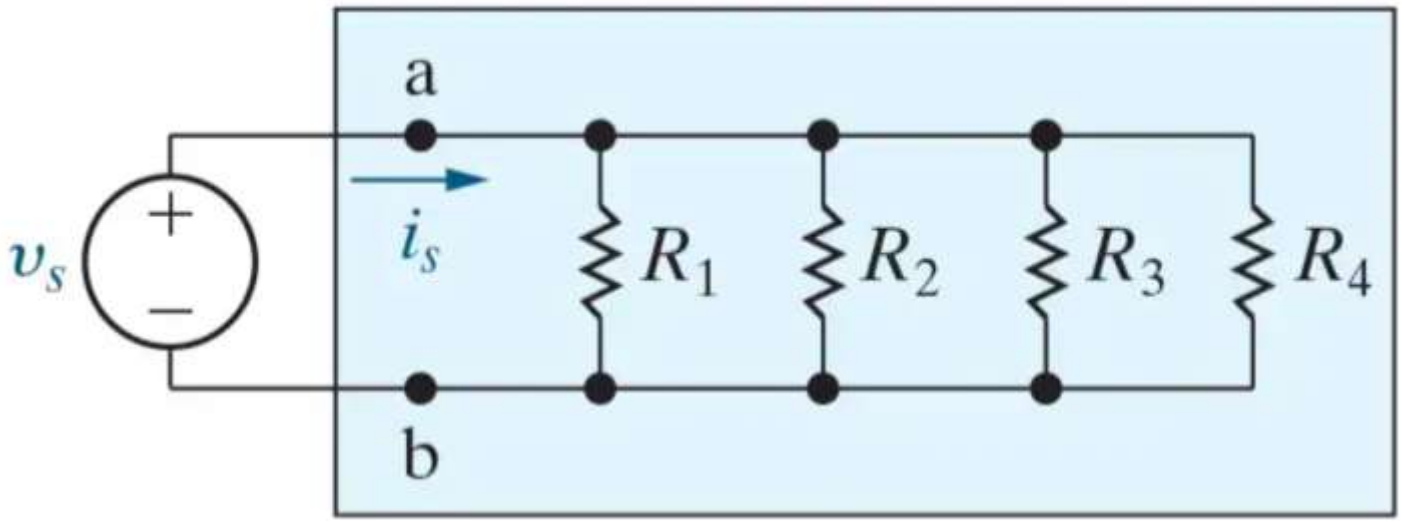
Resistors in parallel:

- Connected at both ends
- Have the same voltage (because of KVL)



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$$i_s = i_1 + i_2 + i_3 + i_4.$$

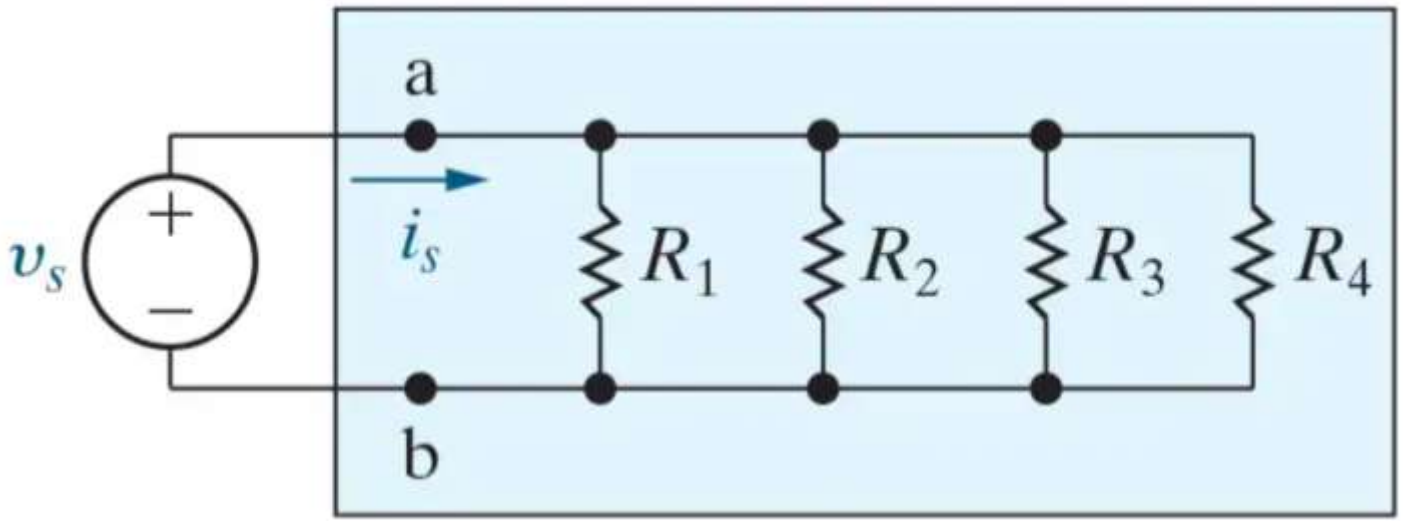
$$i_1 R_1 = i_2 R_2 = i_3 R_3 = i_4 R_4 = v_s.$$

$$i_1 = \frac{v_s}{R_1}, \quad i_2 = \frac{v_s}{R_2}, \quad i_3 = \frac{v_s}{R_3}, \quad \text{and} \quad i_4 = \frac{v_s}{R_4}.$$

$$i_s = v_s \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right),$$

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$$i_s = i_1 + i_2 + i_3 + i_4.$$

$$i_1 R_1 = i_2 R_2 = i_3 R_3 = i_4 R_4 = v_s.$$

$$i_1 = \frac{v_s}{R_1}, \quad i_2 = \frac{v_s}{R_2}, \quad i_3 = \frac{v_s}{R_3}, \quad \text{and} \quad i_4 = \frac{v_s}{R_4}.$$

$$i_s = v_s \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right),$$

$$\frac{i_s}{v_s} = \frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}.$$

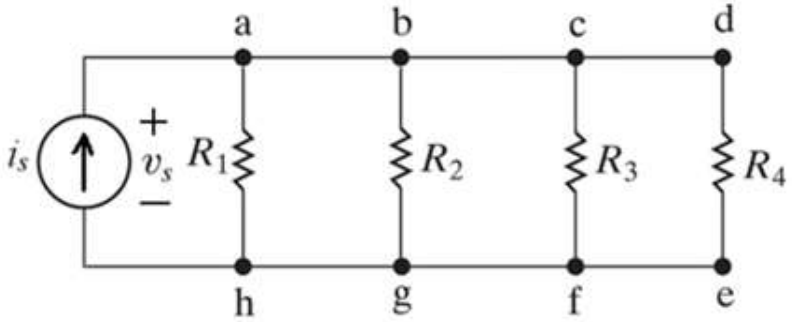
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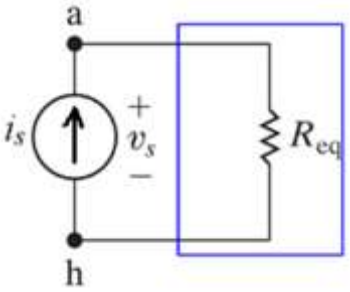
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Chapter 3 – Simple resistive circuits
(9 of 20)

Resistors in parallel:

- Find the equivalent resistance





KCL: $-i_s + v_s/R_1 + v_s/R_2 + \cdots + v_s/R_4 = 0$

$\Rightarrow \quad i_s = (1/R_1 + 1/R_2 + \cdots + 1/R_4)v_s$

KCL: $-i_s + v_s/R_{eq} = 0$

$\Rightarrow \quad i_s = v_s/(R_{eq})$

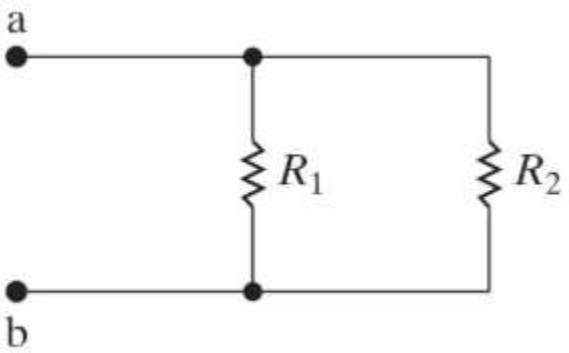
Resistors in parallel: $1/R_{eq} = (1/R_1 + 1/R_2 + \cdots + 1/R_4)$

Equivalent resistance is always smaller than the smallest resistance in the group.

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Chapter 3 – Simple resistive circuits



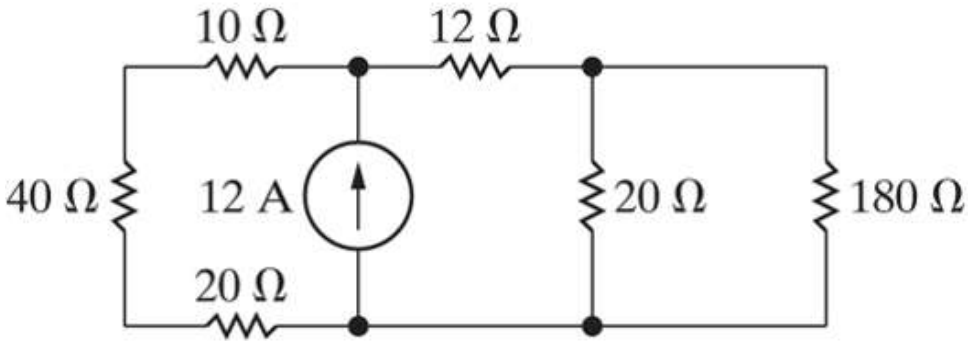
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}.$$

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Chapter 3 – Simple resistive circuits
(10 of 20)

Which of the resistors in the circuit below are connected in parallel?



A. 10Ω and 12Ω

B. 20Ω and 180Ω

C. 10Ω and 40Ω and 20Ω

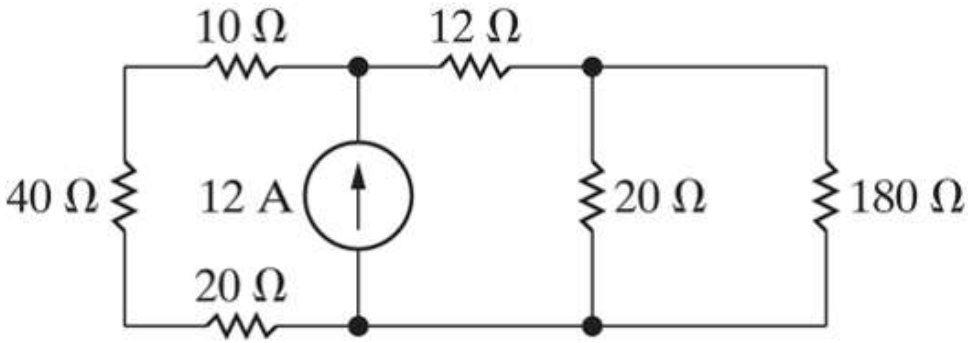
D. 12Ω and 180Ω

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Chapter 3 – Simple resistive circuits
(10 of 20)

Which of the resistors in the circuit below are connected in parallel?



✗ A. 10Ω and 12Ω

✓ B. 20Ω and 180Ω

✗ C. 10Ω and 40Ω and 20Ω

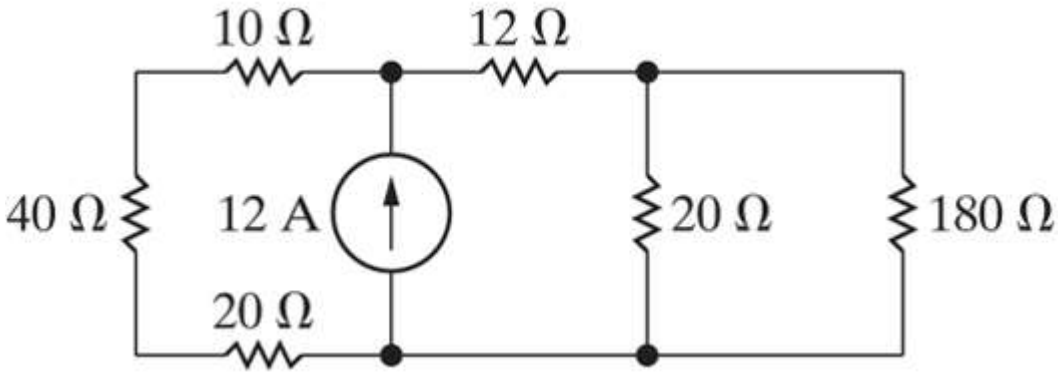
✗ D. 12Ω and 180Ω

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Chapter 3 – Simple resistive circuits
(11 of 20)

Find the equivalent resistance of the parallel-connected resistors in the circuit here



A. 70 Ω

B. 200 Ω

C. 82 Ω

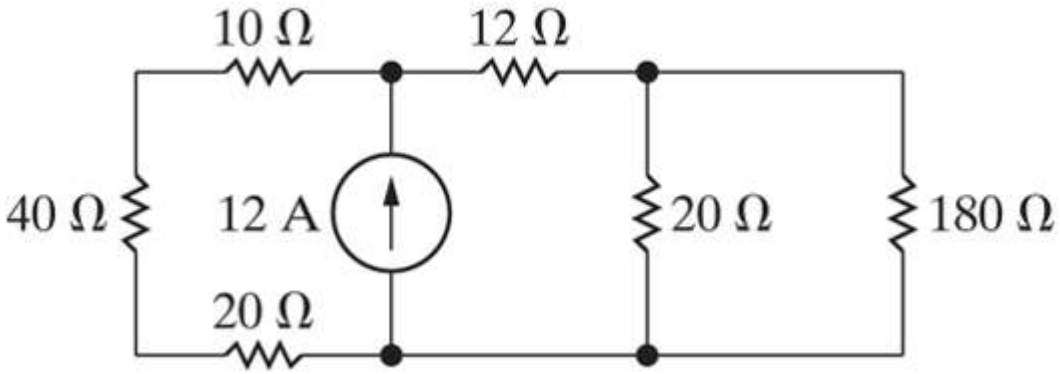
D. 18 Ω

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Chapter 3 – Simple resistive circuits
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Find the equivalent resistance of the parallel-connected resistors in the circuit here



- ✗ A. 70 Ω
- ✗ B. 200 Ω
- ✗ C. 82 Ω
- ✓ D. 18 Ω

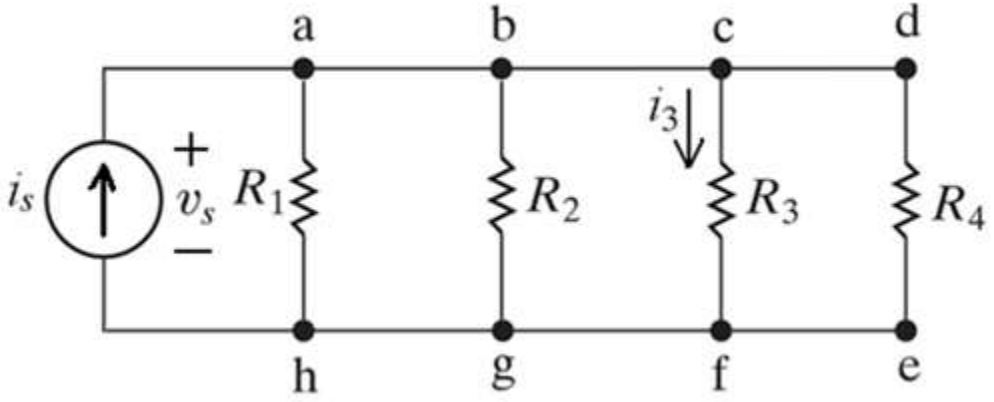
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Chapter 3 – Simple resistive circuits
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Current division:

- Find the current through one of a group of parallel-connected resistors



$$i_3 = \frac{v_s}{R_3} = \frac{R_{eq} i_s}{R_3} = \frac{R_{eq}}{R_3} i_s$$

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Chapter 3 – Simple resistive circuits
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Duality

Voltage sources	Current sources
Resistance	Conductance
KVL	KCL
Series connection	Parallel connection
Voltage division	Current division
Short circuit	Open circuit
…	…

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Chapter 3 – Simple resistive circuits
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Review:

- Series-connected components
- Resistors in series add

$$R_{eq} = \sum_{i=1}^{n} R_i$$

- Voltage division allows you to find the voltage across one component when you know the voltage across a collection of series-connected components:

$$v_i = \frac{R_i}{R_{eq}} v_s$$

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Chapter 3 – Simple resistive circuits
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Review:

- Parallel-connected components
- Add the inverse resistance of resistors in parallel to get the inverse of the equivalent resistance

$$\frac{1}{R_{eq}} = \sum_{i=1}^{n} \frac{1}{R_i}$$

- Current division allows you to find the current through one component when you know the current supplied to a collection of parallel-connected components:

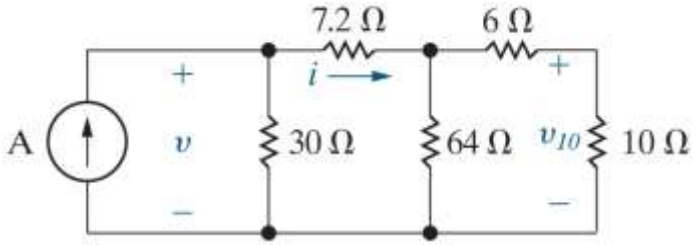
$$i_x = \frac{R_{eq}}{R_x} i_s$$

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Chapter 3 – Simple resistive circuits
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Example: Find *v, i,* and v_{10} by combining resistors in series and parallel, voltage division, and current division.

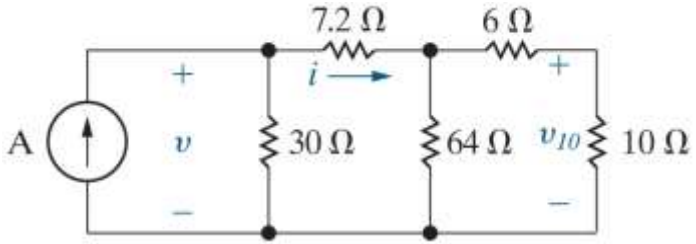


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Chapter 3 – Simple resistive circuits
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Example: Find *v*, *i*, and v_{10} by combining resistors in series and parallel, voltage division, and current division.



Find the equivalent resistance:

$12.8 + 7.2 = 20\Omega$

$6 + 10 = 16\Omega$

$30 \| 20 = 12\Omega$

$64 \| 16 = 12.8\Omega$

Ohm's Law: $v = R_{eq}(5) = (12)(5) = 60 \text{ V}$

KCL: $i = \frac{R_{eq}}{R}(5) = \frac{30 \| 20}{20}(5) = 3 \text{ A}$

Ohm's Law: $v' = i(64 \| 16) = (3)(12.8) = 38.4 \text{ V}$

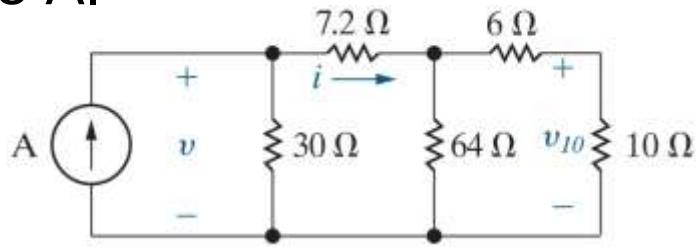
KVL: $v_{10} = \frac{10}{10+6}v' = \frac{10}{16}(38.4) = 24 \text{ V}$

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Chapter 3 – Simple resistive circuits (17 of 20)

Example: Find *v*, *i*, and v_{10} by combining resistors in series and parallel, voltage division, and current division. The current source operates at 5 A.



Find the equivalent resistance:

$12.8 + 7.2 = 20\Omega$

$6 + 10 = 16\Omega$

$30 \| 20 = 12\Omega$

$64 \| 16 = 12.8\Omega$

Ohm's Law: $v = R_{eq}(5) = (12)(5) = 60 \text{ V}$

KCL: $i = \frac{R_{eq}}{R}(5) = \frac{30 \| 20}{20}(5) = 3 \text{ A}$

Ohm's Law: $v' = i(64 \| 16) = (3)(12.8) = 38.4 \text{ V}$

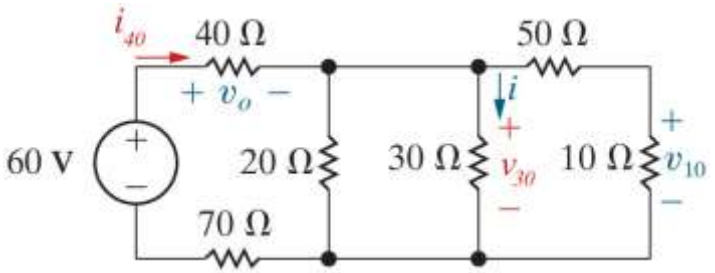
KVL: $v_{10} = \frac{10}{10+6}v' = \frac{10}{16}(38.4) = 24 \text{ V}$

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Chapter 3 – Simple resistive circuits
(20 of 20)

Example: Find v, i, and v_{10} using series and parallel simplification, and voltage and current division.

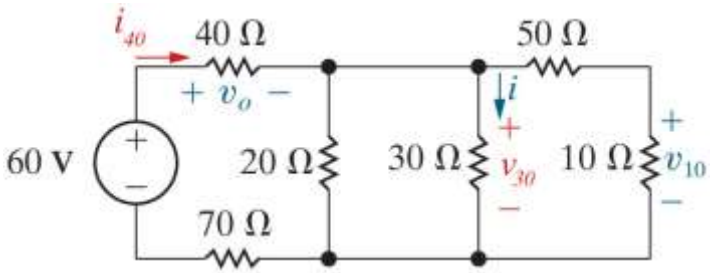


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Chapter 3 – Simple resistive circuits
(20 of 20)

Example: Find v, i, and v_{10} using series and parallel simplification, and voltage and current division.



$$20 \,||\, 30 \,||\, 60 = 10\,\Omega \qquad 50 + 10 = 60\,\Omega$$

voltage division: $v_o = \dfrac{40}{40+10+70}(60) = 20 \text{ V}$

Ohm's Law: $i_{40} = \dfrac{v_o}{40} = \dfrac{20}{40} = 0.5 \text{ A}$

current division: $i = \dfrac{20 \,||\, 30 \,||\, 60}{30} i_{40} = \dfrac{10}{30}(0.5) = 0.167 \text{ A}$

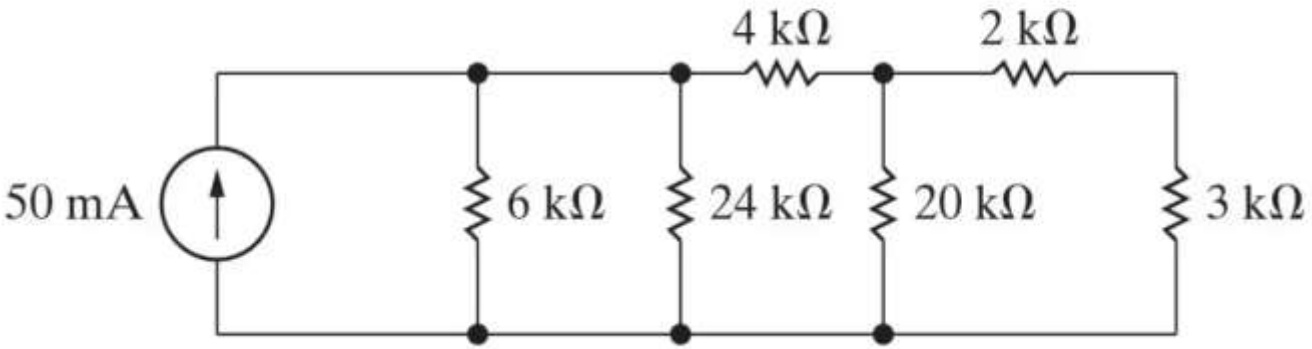
Ohm's Law: $v_{30} = 30i = 30(0.167) = 5 \text{ V}$

voltage division: $v_{10} = \dfrac{10}{10+50} v_{30} = \dfrac{10}{60}(5) = 0.833 \text{ V}$

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Chapter 3 – Simple resistive circuits



Find the equivalent resistance seen by the current source

Find the power delivered by the current source

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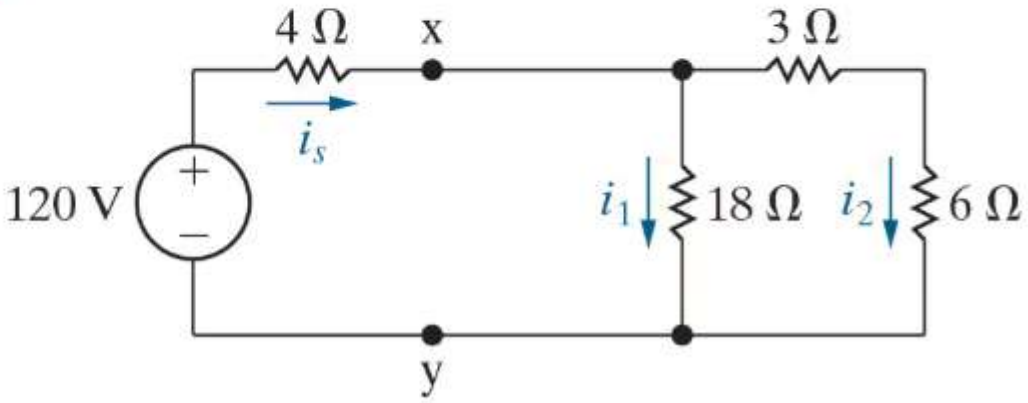
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Chapter 3 – Simple resistive circuits

Find i_s, i_1, and i_2 in the circuit shown in Fig. 3.11.

Figure 3.11

The circuit for Example 3.2.

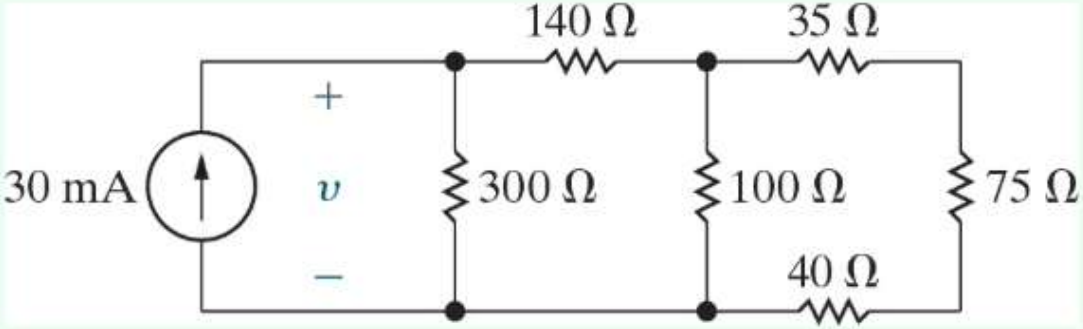


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Chapter 3 – Simple resistive circuits

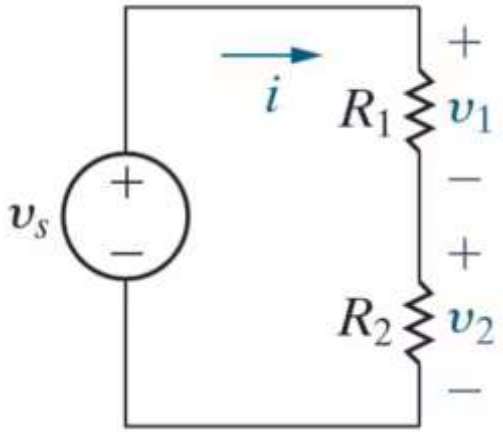
3.1 For the circuit shown, find (a) the voltage v, (b) the power delivered to the circuit by the current source, and (c) the power dissipated in the 75 Ω resistor.



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Chapter 3 – Simple resistive circuits

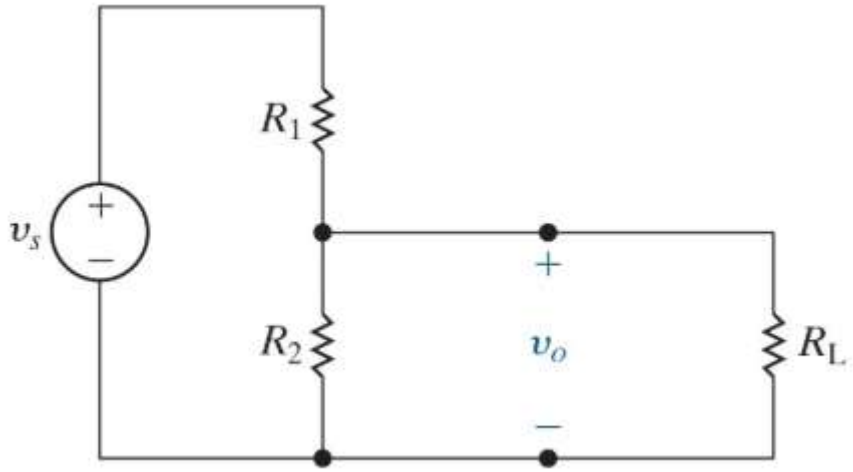


$$v_1 = iR_1 = v_s \frac{R_1}{R_1 + R_2},$$

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$$v_o = \frac{R_{eq}}{R_1 + R_{eq}} v_s,$$

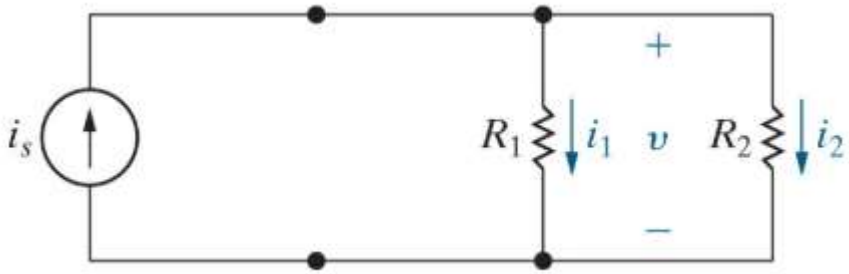
$$R_{eq} = \frac{R_2 R_L}{R_2 + R_L}.$$

$$v_o = \frac{R_2}{R_1[1 + (R_2/R_L)] + R_2} v_s.$$

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$$i_1 = \frac{R_2}{R_1 + R_2} i_s,$$

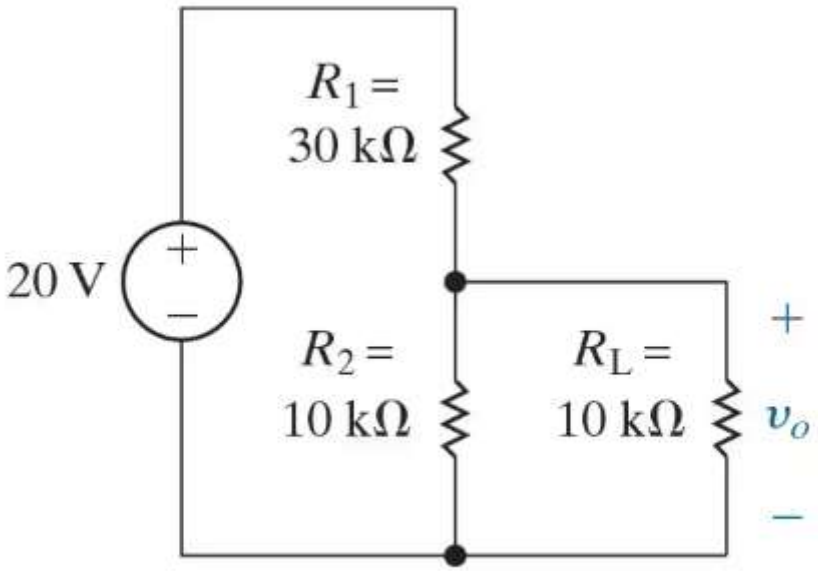
$$v = i_1 R_1 = i_2 R_2 = \frac{R_1 R_2}{R_1 + R_2} i_s.$$

$$i_2 = \frac{R_1}{R_1 + R_2} i_s.$$

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a. For the voltage divider designed in Example 3.3, suppose $R_2 = 10\ \text{k}\Omega$ and $R_1 = 30\ \text{k}\Omega$. Connect a resistor $R_L = 10\ \text{k}\Omega$ in parallel with R_2 and determine the voltage across R_L.

b. Repeat part (a) using $R_2 = 400\ \Omega$ and $R_1 = 1200\ \Omega$, but the same value of R_L.

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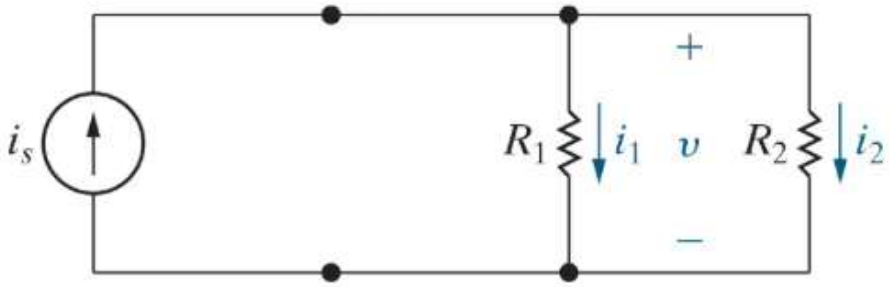
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Suppose the current source for the current divider shown in Fig. 3.19 is 100 mA. Assuming you have 0.25 W resistors available, what is the largest R_2 resistor you can use to get $i_2 = 50$ mA?

Figure 3.19

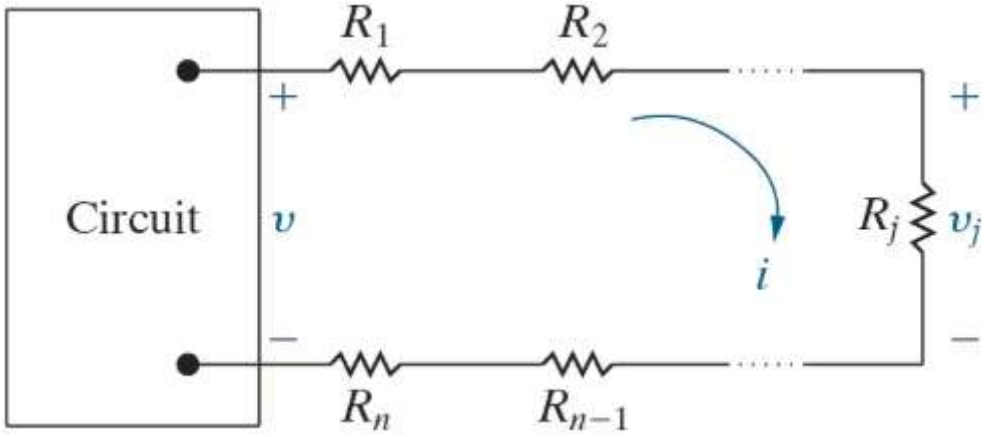
The current-divider circuit.



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Voltage Division Equation

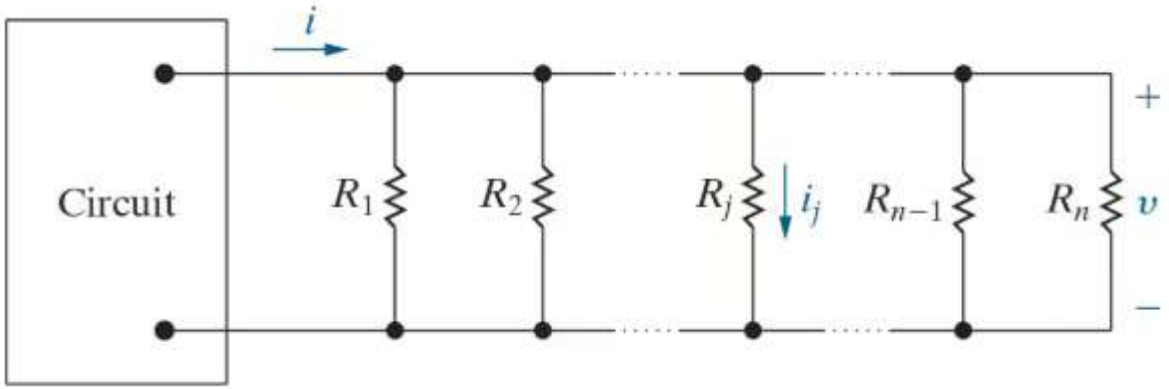
(Equation 3.9)

$$v_j = iR_j = \frac{R_j}{R_{\text{eq}}} v.$$

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Current Division Equation

(Equation 3.10)

$$i_j = \frac{v}{R_j} = \frac{R_{\text{eq}}}{R_j} i.$$

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Example 3.7: Using Voltage Division and Current Division to Solve a Circuit

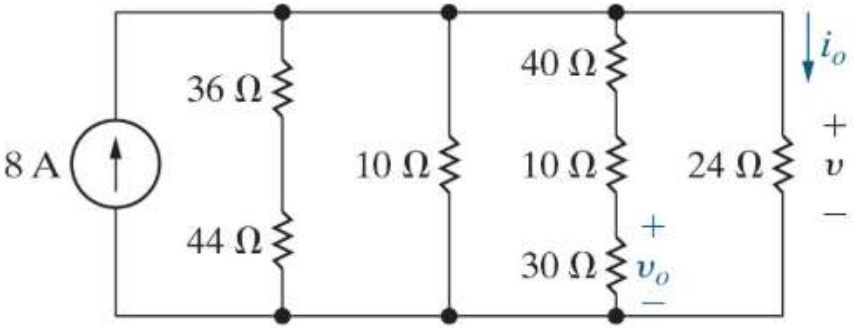
Use current division to find the current i_o and use voltage division to find the voltage v_o for the circuit in Fig. 3.22.

Solution

We can use Eq. 3.10 if we can find the equivalent resistance of the four parallel branches containing resistors. Using "+" to represent series-connected resistors and "||" to represent parallel-connected resistors, the equivalent resistance is

Figure 3.22

The circuit for Example 3.7.



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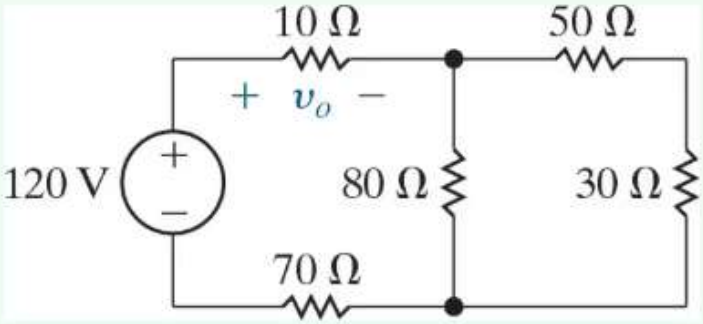
Assessment Problem

Objective 3—Be able to use voltage and current division to solve simple circuits

3.4 **a.** Use voltage division to determine the voltage v_o across the 10 Ω resistor in the circuit shown.

b. Use current division to calculate the current in the 80 Ω resistor. (*Hint:* Start by using v_o from part (a) to find the current in the 10 Ω resistor.)

c. How much power is absorbed by the 50 Ω resistor?



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